

NATAL, Brazil

WMO: 825990

Lat: 5.911S

Lon: 35.248W

Elev: 52

StdP: 100.71

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	21.1	17.8	12.9	26.6	18.2	13.2	26.3	11.0	28.2	10.3	28.2	2.9	180	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.7	32.7	25.4	32.1	25.2	31.8	25.1	26.7	29.9	26.3	29.7	26.1	29.5	6.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.5	28.4	25.2	20.5	27.9	25.1	20.3	27.7	84.0	30.3	82.4	29.9	81.0	29.9	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.5	8.7	19.8	33.4	0.9	1.5	19.2	34.5	18.6	35.4	18.1	36.2	17.5	37.3
			18.6	27.7	1.0	1.1	17.9	28.5	17.3	29.1	16.8	29.8	16.1	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	27.6	27.7	27.8	27.4	26.9	25.9	25.4	25.4	25.9	26.8	27.2	27.7	(6)
	DBStd	1.31	0.98	0.91	0.97	0.97	0.96	1.07	0.96	0.94	1.01	0.99	0.89	0.87	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6135	547	497	552	523	524	477	476	477	477	521	517	547	(10)
	CDD18.3	3093	288	263	293	273	266	227	218	219	227	263	267	289	(11)
	CDH23.3	28747	3059	2836	3141	2711	2386	1656	1431	1538	1829	2458	2666	3038	(12)
	CDH26.7	8976	1012	976	1104	893	744	442	343	397	510	736	823	995	(13)
(14) Wind	WSAvg	4.5	4.5	4.3	3.9	3.7	3.8	4.1	4.6	5.0	5.2	5.4	5.1	4.9	(14)
(15) Precipitation	PrecAvg	1306	54	107	184	201	180	195	176	86	54	23	21	28	(15)
	PrecMax	2116	189	271	401	480	350	427	430	241	164	138	118	110	(16)
	PrecMin	742	1	3	28	57	59	57	41	12	3	1	0	0	(17)
	PrecStd	386	52	84	99	115	84	113	85	61	38	29	27	28	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.0	33.1	33.0	32.2	31.5	31.1	31.2	32.0	32.2	32.6	32.9	(19)
		MCWB	25.4	25.8	25.9	25.9	25.6	25.1	24.4	24.1	24.3	24.2	24.7	25.4	(20)
	2%	DB	32.0	32.1	32.2	32.1	31.6	30.8	30.1	30.2	30.9	31.2	31.8	32.0	(21)
		MCWB	25.2	25.3	25.5	25.6	25.4	24.8	23.9	23.6	24.0	24.2	24.6	25.0	(22)
	5%	DB	31.2	31.2	31.7	31.2	31.0	30.0	29.2	29.5	30.0	30.4	30.9	31.1	(23)
		MCWB	24.9	25.0	25.3	25.4	25.2	24.4	23.6	23.3	23.6	23.8	24.3	24.7	(24)
	10%	DB	30.7	30.9	31.0	30.8	30.1	29.1	28.8	28.9	29.2	29.9	30.1	30.2	(25)
		MCWB	24.7	25.0	25.2	25.3	25.0	24.1	23.4	23.0	23.2	23.6	23.9	24.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.8	26.8	27.1	27.1	26.4	25.7	25.5	25.5	25.6	25.9	26.2	(27)
		MCDB	30.1	30.2	30.5	30.1	29.8	29.1	28.7	28.2	28.7	29.7	30.1	30.2	(28)
	2%	WB	26.2	26.2	26.3	26.6	26.5	25.7	25.1	24.8	24.9	24.9	25.3	25.7	(29)
		MCDB	29.7	29.8	29.9	29.8	29.5	28.7	28.1	28.1	28.6	29.5	29.8	30.0	(30)
	5%	WB	25.8	25.9	26.1	26.2	26.1	25.3	24.6	24.4	24.5	24.5	24.9	25.4	(31)
		MCDB	29.4	29.5	29.7	29.5	29.2	28.3	27.6	27.7	28.2	28.9	29.4	29.6	(32)
	10%	WB	25.4	25.6	25.7	25.9	25.7	24.8	24.2	23.9	24.1	24.2	24.5	25.0	(33)
		MCDB	29.0	29.2	29.3	29.2	28.7	27.7	27.1	27.2	27.8	28.3	28.8	29.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.7	6.7	6.7	6.8	6.8	7.0	7.6	7.4	6.8	6.5	6.4	(35)
		MCDBR	7.0	7.1	7.4	7.4	7.7	7.7	7.8	8.2	8.0	7.2	7.1	6.9	(36)
	5% WB	MCWBR	2.4	2.4	2.5	2.7	2.9	3.0	3.2	3.1	2.8	2.4	2.3	2.3	(37)
		MCWBR	6.3	6.7	6.9	6.9	6.9	6.9	7.0	7.4	7.4	6.9	6.7	6.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.401	0.406	0.399	0.407	0.391	0.392	0.393	0.398	0.420	0.401	0.403	0.412	(40)
		taud	2.486	2.477	2.514	2.506	2.545	2.537	2.518	2.469	2.379	2.437	2.439	2.433	(41)
	Ebn at Noon		937	929	924	891	878	862	868	886	890	923	928	922	(42)
		Edh at Noon	116	117	111	108	99	98	101	110	125	120	120	121	(43)
(44) All-Sky Solar Radiation	RadAvg		6.02	6.03	5.88	5.31	5.03	4.56	4.71	5.54	6.07	6.43	6.55	6.24	(44)
	RadStd		0.40	0.33	0.30	0.34	0.34	0.25	0.32	0.19	0.27	0.22	0.20	0.22	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-0.48	N/A	-0.31	-0.30	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	-0.48	N/A	-0.31	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page